CO1_AT3_Visualization Design Exercise

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DSA0601- Data Handling and Visualization

Question 1

A multinational retail company has collected annual sales information from its branches across different regions. Using the given dataset, design an effective visualization that simultaneously represents **Sales, Profit, Number of Customers, Product Category, and Region**. Select suitable aesthetic mappings and apply an appropriate scale transformation wherever necessary. Justify every visualization design choice.

R Dataset

retail <- data.frame(

Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",

"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),

Region=c("North","South","East","West","North","South","East","West

", "North","South","East","West","North","South","East","West",

"North","South","East","West"),

Category=c("Electronics","Furniture","Clothing","Food","Electronics",

"Furniture","Clothing","Food","Electronics","Furniture",

"Clothing","Food","Electronics","Furniture","Clothing",

"Food","Electronics","Furniture","Clothing","Food"),

Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,

560000,670000,195000,980000,430000,760000,280000,1150000,

510000,690000,220000,1250000),

Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000, 72000,98000,-8000,205000,51000,132000,24000,250000,

69000,115000,15000,285000),

Customers=c(1200,2500,980,4200,1800,3000,1250,5200,

2100,3300,1100,4700,1750,3600,1450,5600,

2200,3900,1600,6100)
)

Aim

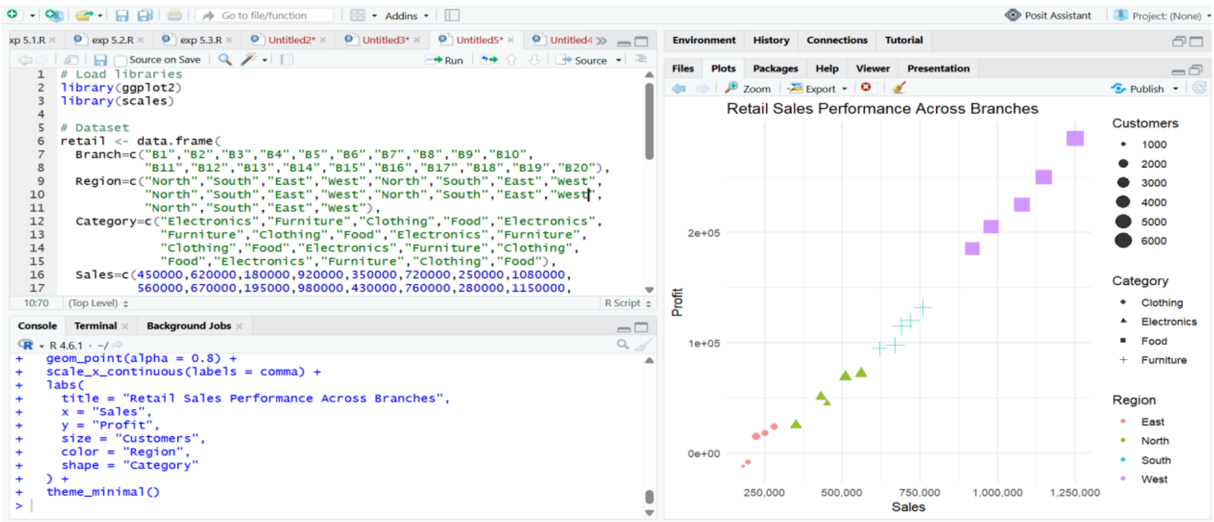
Design a bubble chart to visualize Sales, Profit, Customers, Product Category, and Region of retail branches using **ggplot2**.

R Code

```
# Install packages (Run only once)
install.packages("ggplot2")
install.packages("scales")

# Load libraries
library(ggplot2)
library(scales)

# Dataset
retail <- data.frame(
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",

"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),

Region=c("North","South","East","West","North","South","East","West"
,

"North","South","East","West","North","South","East","West",
           "North","South","East","West"),

Category=c("Electronics","Furniture","Clothing","Food","Electronics"
,

"Furniture","Clothing","Food","Electronics","Furniture",
             "Clothing","Food","Electronics","Furniture","Clothing",
             "Food","Electronics","Furniture","Clothing","Food"),
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
          560000,670000,195000,980000,430000,760000,280000,1150000,
          510000,690000,220000,1250000),
  Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
          72000,98000,-8000,205000,51000,132000,24000,250000,
          69000,115000,15000,285000),
  Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
              2100,3300,1100,4700,1750,3600,1450,5600,
              2200,3900,1600,6100)
)
```

```
# Bubble Chart
ggplot(retail,
       aes(x = Sales,
           y = Profit,
           size = Customers,
           color = Region,
           shape = Category)) +
  geom_point(alpha = 0.8) +
  scale_x_continuous(labels = comma) +
  labs(
    title = "Retail Sales Performance Across Branches",
    x = "Sales",
    y = "Profit",
    size = "Customers",
    color = "Region",
    shape = "Category"
  ) +
  theme_minimal()
```

OUTPUT:



Question 2

The following dataset contains monthly electricity generation from different energy sources. Develop visualizations using **Cartesian Coordinates** and **Polar Coordinates**. Compare both visualizations and determine which coordinate system provides better analytical insight. Explain your design decisions.

R Dataset

energy <- data.frame(

Month=month.abb,

Coal=c(620,580,590,605,640,670,690,720,700,680,650,630), Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),

Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),

Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),

Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)

)

Aim

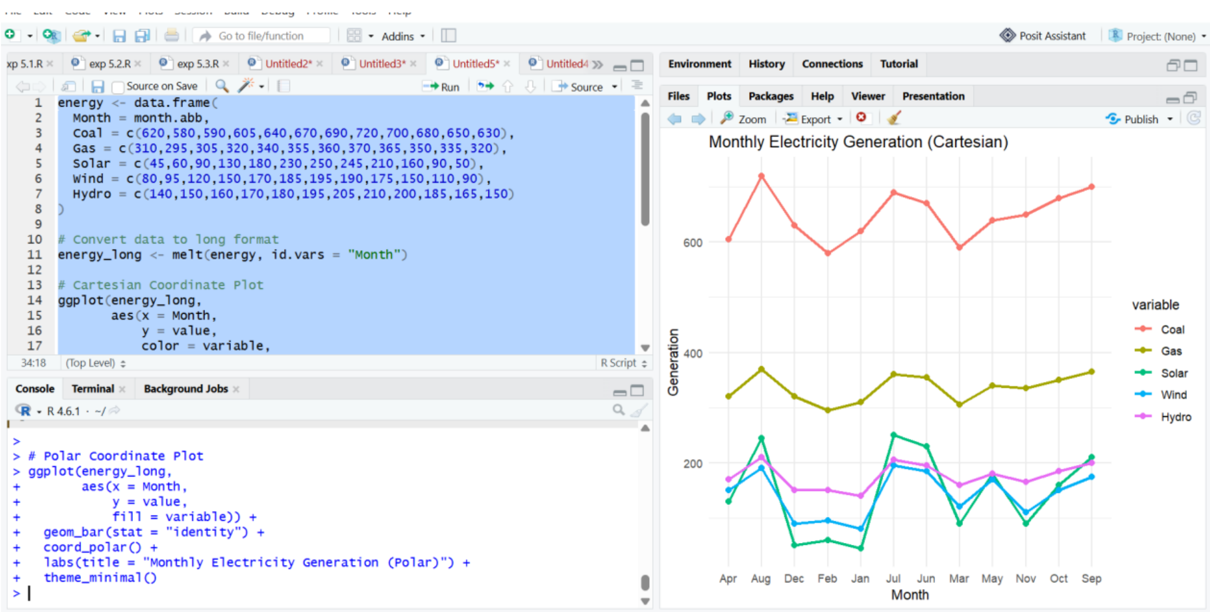
Develop Cartesian and Polar coordinate visualizations to compare monthly electricity generation from different energy sources.

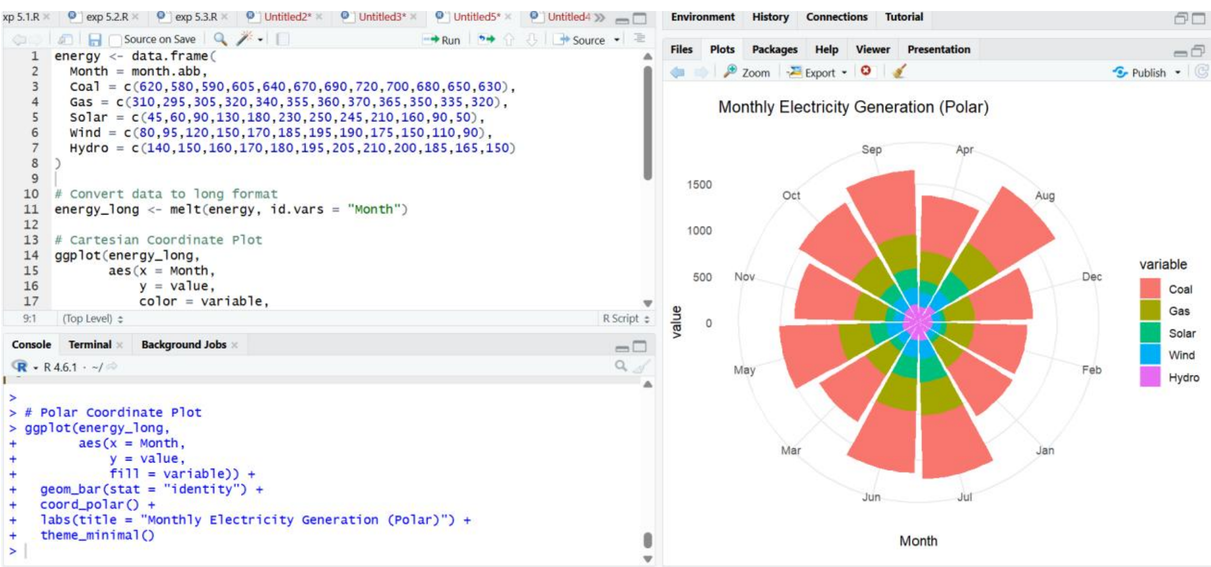
R Code

```
# Install packages (Run only once)
install.packages("ggplot2")
install.packages("reshape2")

# Load libraries
library(ggplot2)
library(reshape2)

# Dataset
energy <- data.frame(
  Month = month.abb,
  Coal = c(620,580,590,605,640,670,690,720,700,680,650,630),
  Gas = c(310,295,305,320,340,355,360,370,365,350,335,320),
  Solar = c(45,60,90,130,180,230,250,245,210,160,90,50),
  Wind = c(80,95,120,150,170,185,195,190,175,150,110,90),
  Hydro = c(140,150,160,170,180,195,205,210,200,185,165,150)
)

# Convert data to long format
energy_long <- melt(energy, id.vars = "Month")

# Cartesian Coordinate Plot
ggplot(energy_long,
       aes(x = Month,
           y = value,
           color = variable,
           group = variable)) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  labs(title = "Monthly Electricity Generation (Cartesian)",
       x = "Month",
       y = "Generation") +
  theme_minimal()
```

```
# Polar Coordinate Plot
ggplot(energy_long,
       aes(x = Month,
           y = value,
           fill = variable)) +
  geom_bar(stat = "identity") +
  coord_polar() +
  labs(title = "Monthly Electricity Generation (Polar)") +
  theme_minimal()
```

Output





Question 3

An environmental monitoring agency has collected climate data from major Indian cities. Design a visualization that effectively represents **Temperature, Air Quality Index (AQI), Rainfall, and Humidity** using appropriate colour scales. Select a suitable colour palette and justify its effectiveness for accurate interpretation and accessibility.

R Dataset

```
climate <- data.frame(

City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",

"Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",

"Bhopal","Patna","Goa","Kochi","Nagpur","Indore",

"Surat","Visakhapatnam","Coimbatore","Mysuru"),
Temperature=c(44,35,38,40,39,31,33,43,45,39,

38,40,32,30,41,37,36,34,29,28),

Humidity=c(22,84,78,72,48,66,58,26,18,42,

36,45,89,91,30,34,76,82,68,64),

Rainfall=c(6,280,220,190,85,120,95,2,1,35,

48,72,360,410,12,20,180,240,130,145),

AQI=c(365,110,125,180,160,72,84,290,320,240,

175,220,58,52,270,190,145,98,60,55)

)
```

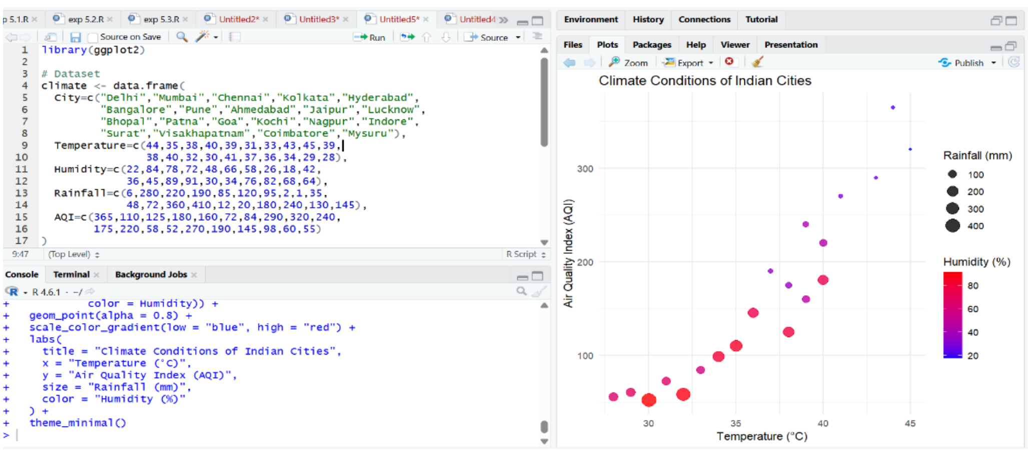
Aim

Design a visualization to represent Temperature, Humidity, Rainfall, and Air Quality Index (AQI) of Indian cities using appropriate colour scales.

R Code

```
# Install package (Run only once)
install.packages("ggplot2")

# Load library
library(ggplot2)

# Dataset
climate <- data.frame(
  City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",
         "Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",
         "Bhopal","Patna","Goa","Kochi","Nagpur","Indore",
         "Surat","Visakhapatnam","Coimbatore","Mysuru"),
  Temperature=c(44,35,38,40,39,31,33,43,45,39,
                38,40,32,30,41,37,36,34,29,28),
  Humidity=c(22,84,78,72,48,66,58,26,18,42,
             36,45,89,91,30,34,76,82,68,64),
  Rainfall=c(6,280,220,190,85,120,95,2,1,35,
             48,72,360,410,12,20,180,240,130,145),
  AQI=c(365,110,125,180,160,72,84,290,320,240,
        175,220,58,52,270,190,145,98,60,55)
)

# Bubble Chart
ggplot(climate,
       aes(x = Temperature,
           y = AQI,
           size = Rainfall,
           color = Humidity)) +
  geom_point(alpha = 0.8) +
  scale_color_gradient(low = "blue", high = "red") +
  labs(
    title = "Climate Conditions of Indian Cities",
    x = "Temperature (°C)",
    y = "Air Quality Index (AQI)",
    size = "Rainfall (mm)",
    color = "Humidity (%)"
  ) +
  theme_minimal()
```

Output



Question 4

A software company has recorded project statistics for twenty software development teams. Design a visualization that requires the use of a **nonlinear axis transformation** to represent highly skewed data. Justify the selected transformation and explain how it improves interpretation over a linear scale.

R Dataset

```
software <- data.frame(

Team=paste("T",1:20,sep=""),

LinesOfCode=c(12000,18000,25000,35000,48000,

62000,81000,96000,120000,150000,

185000,240000,310000,390000,480000,

620000,810000,1050000,1350000,1800000),

Bugs=c(35,52,60,72,88,

110,125,142,165,188,

205,240,280,320,365,

420,490,560,640,720),

Developers=c(5,7,9,10,12,

14,16,18,20,22,
25,28,30,34,38,

42,46,50,55,60),

Complexity=c(2.1,2.4,2.7,3.0,3.2,

3.5,3.8,4.1,4.5,4.8,

5.1,5.4,5.8,6.2,6.6,
```

7.1,7.6,8.0,8.5,9.0)

)

Aim

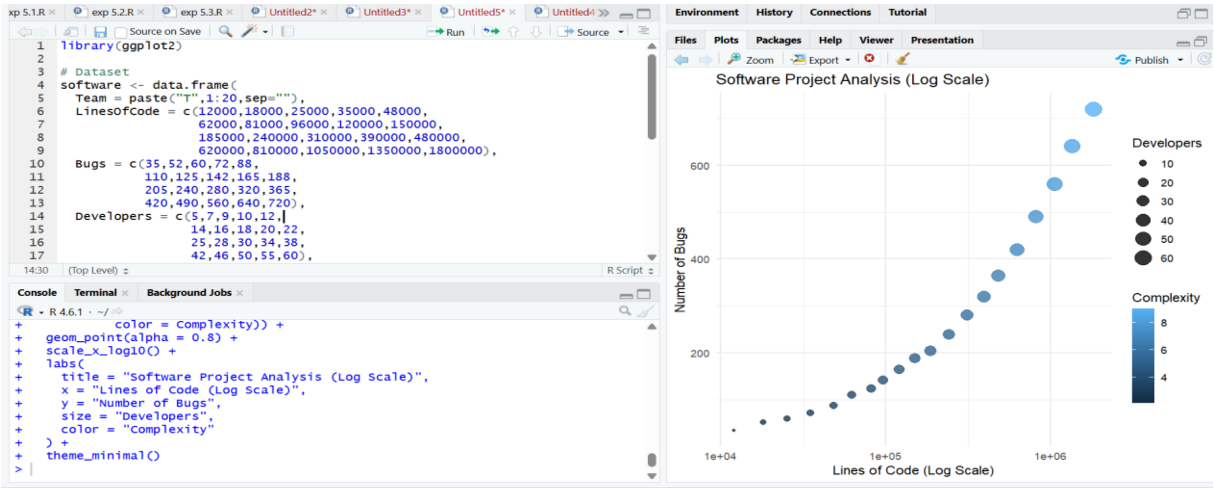
Design a visualization using a nonlinear (logarithmic) axis transformation to represent highly skewed software project data.

R Code

```
# Install package (Run only once)
install.packages("ggplot2")

# Load library
library(ggplot2)

# Dataset
software <- data.frame(
  Team = paste("T",1:20,sep=""),
  LinesOfCode = c(12000,18000,25000,35000,48000,
                  62000,81000,96000,120000,150000,
                  185000,240000,310000,390000,480000,
                  620000,810000,1050000,1350000,1800000),
  Bugs = c(35,52,60,72,88,
           110,125,142,165,188,
           205,240,280,320,365,
           420,490,560,640,720),
  Developers = c(5,7,9,10,12,
                 14,16,18,20,22,
                 25,28,30,34,38,
                 42,46,50,55,60),
  Complexity = c(2.1,2.4,2.7,3.0,3.2,
                 3.5,3.8,4.1,4.5,4.8,
                 5.1,5.4,5.8,6.2,6.6,
                 7.1,7.6,8.0,8.5,9.0)
)

# Scatter Plot with Log Scale
ggplot(software,
       aes(x = LinesOfCode,
           y = Bugs,
           size = Developers,
           color = Complexity)) +
  geom_point(alpha = 0.8) +
  scale_x_log10() +
  labs(
    title = "Software Project Analysis (Log Scale)",
    x = "Lines of Code (Log Scale)",
    y = "Number of Bugs",
    size = "Developers",
    color = "Complexity"
```

```
) +
theme_minimal()
```

Output



Question 5

A financial analytics company has collected performance data for twenty publicly listed companies. Develop an integrated visualization that effectively communicates **Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector**. Use appropriate aesthetic mappings, coordinate systems, scales, and colour palettes to maximise interpretability while minimising perceptual bias.

R Dataset

companies <- data.frame(

Company=paste("C",1:20,sep=""),

Sector=c("IT","Banking","Healthcare","Retail","Energy",

"IT","Banking","Healthcare","Retail","Energy",

"IT","Banking","Healthcare","Retail","Energy",

"IT","Banking","Healthcare","Retail","Energy"),

Revenue=c(420,580,720,850,980,

1150,1320,1480,1650,1820,

2050,2280,2550,2840,3150,

3480,3860,4250,4680,5200),

ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,

20.4,17.8,14.2,10.5,22.1,

24.5,19.8,16.7,13.4,26.2,
28.1,21.5,18.3,15.2,30.4),

Employees=c(1500,3200,2100,4200,2800
, 5100,6800,3900,7200,4500,
8100,9500,6200,10300,7200,

11800,13200,8600,14500,16000),
MarketCap=c(8500,12000,9800,7600,14500
, 18200,21500,16800,13200,24500,
28500,32500,24800,19200,38200,
42500,46800,35600,29800,52000))

Aim

To design an integrated bubble chart that represents Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector using appropriate aesthetic mappings, colour scales, and coordinate systems.

R Code

```
# Install package (Run only once)
install.packages("ggplot2")

# Load library
library(ggplot2)

# Dataset
companies <- data.frame(
  Company = paste("C",1:20,sep=""),
  Sector = c("IT","Banking","Healthcare","Retail","Energy",
             "IT","Banking","Healthcare","Retail","Energy",
             "IT","Banking","Healthcare","Retail","Energy",
             "IT","Banking","Healthcare","Retail","Energy"),
```

```
  Revenue = c(420,580,720,850,980,
              1150,1320,1480,1650,1820,
              2050,2280,2550,2840,3150,
              3480,3860,4250,4680,5200),
  ProfitMargin = c(8.2,15.6,11.4,9.3,18.5,
                   20.4,17.8,14.2,10.5,22.1,
                   24.5,19.8,16.7,13.4,26.2,
                   28.1,21.5,18.3,15.2,30.4),
  Employees = c(1500,3200,2100,4200,2800,
                5100,6800,3900,7200,4500,
                8100,9500,6200,10300,7200,
                11800,13200,8600,14500,16000),
  MarketCap = c(8500,12000,9800,7600,14500,
                18200,21500,16800,13200,24500,
                28500,32500,24800,19200,38200,
                42500,46800,35600,29800,52000)
)

# Bubble Chart
ggplot(companies,
       aes(x = Revenue,
           y = ProfitMargin,
           size = MarketCap,
           color = Sector)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Company Financial Performance",
    x = "Revenue",
    y = "Profit Margin (%)",
    size = "Market Capitalization",
    color = "Industry Sector"
  ) +
  theme_minimal()
```

Output: